

CSE211 Digital Design

Akdeniz University

Week13: Sequential Logic Part 2

1

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Course program

2

Week 01	15/09/2025	Introduction
Week 02	22/09/2025	Digital Systems and Binary Numbers I
Week 03	29/09/2025	Digital Systems and Binary Numbers II
Week 04	06/10/2025	Boolean Algebra and Logic Gates I
Week 05	13/10/2025	Boolean Algebra and Logic Gates II
Week 06	20/10/2025	Gate Level Minimization
Week 07	27/10/2025	Karnaugh Maps
Week 08	27/10/2025	Karnaugh Maps
Week 09	05/11/2025	Midterm (Expected)
Week 10	17/11/2025	Combinational Logic
Week 11	24/11/2025	Combinational Logic
Week 12	01/12/2025	Timing, delays and hazards
Week 13	08/12/2025	Synchronous Sequential Logic
Week 14	15/12/2025	Synchronous Sequential Logic

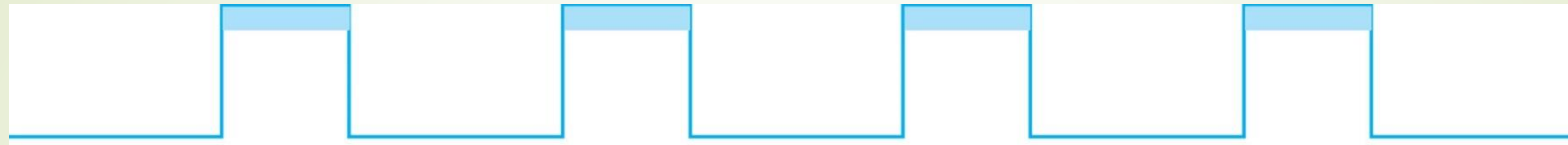
5.4 Storage Elements: Flip-Flops

3

- The state of a latch or flip-flop is switched by a change in the control input. This momentary change is called a **trigger**
- The problem with the latch is that it responds to a change in the *level* of a clock pulse. Glitches are problematic.

Clock response in latch and flip-flop

4



(a) Response to positive level



(b) Positive-edge response



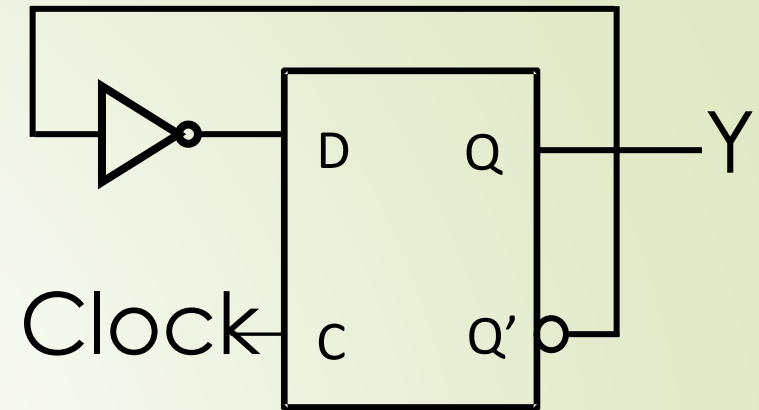
(c) Negative-edge response

Flip-Flops

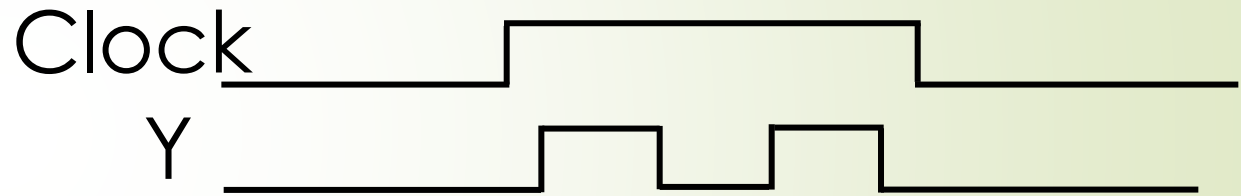
- The latch timing problem
- Edge-triggered flip-flop
- Master-slave flip-flop
- Direct inputs to flip-flops
- Flip-flop timing

The Latch Timing Problem

➤ Consider the following circuit:



➤ Suppose that initially $Y = 0$.



- As long as $C = 1$, the value of Y continues to change!
- The changes are based on the delay present on the loop through the connection from Y back to Y .
- This behavior is clearly unacceptable.
- Desired behavior: Y changes only once per clock pulse

Master-slave D flip-flop

7

- The circuit samples the D input and **changes** its output Q only at the **negative edge** of the synchronizing or controlling clock.
- Another say, a change in the output of the flip-flop can be triggered only by and during the transition of the clock from 1 to 0.

